CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

ENTHUSIAST ADVANCE COURSE

PHASE : MEB-I (A)

TARGET : PRE-MEDICAL 2024

Test Type : **MINOR** Test Pattern : **NEET (UG)**

TEST DATE : 16-07-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	1	2	3	3	2	2	3	4	3	2	1	1	1	2	3	4	1	4	2	2	2	1	3	2	1	1	3	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	4	1	4	2	3	3	4	1	4	2	1	3	3	2	1	4	1	1	2	2	1	4	1	4	2	2	3	3	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	2	4	1	4	2	2	2	3	2	3	3	1	3	4	4	1	3	3	4	1	1	3	3	3	3	2	3	4	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	2	4	4	3	1	4	1	2	2	4	4	3	3	2	4	3	3	4	3	4	3	4	1	3	3	2	2	1	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	1	1	2	3	1	2	2	2	4	3	1	2	3	3	3	3	1	4	2	2	1	2	2	1	2	1	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	1	1	3	3	2	3	3	4	2	2	1	2	1	4	3	3	1	1	4	3	4	3	4	2	2	3	3	3	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	3	1	4	1	3	1	1	2	3	1	3	1	3	3	4	1	2	3	2										

HINT – SHEET

SUBJECT : PHYSICS

SECTION-A

1. Ans (1)

$\lambda = 2[0.49 - 0.16]$

$\lambda = 2[0.33]$

$n = \dfrac{v}{\lambda} = \dfrac{330}{2 \times 0.33} = 500\,\text{Hz}$

2. Ans (1)

Fundamental frequency

$n = \dfrac{1}{2L}\sqrt{\dfrac{T}{m}}$ $\quad [\because\ m = \dfrac{M}{L}] \quad \therefore\ n$

$= \dfrac{1}{2}\sqrt{\dfrac{T}{ML}}$

4. Ans (3)

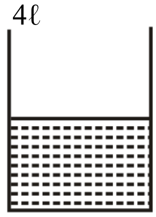
Number of beats $\quad \Delta n = n_1 - n_2$

$2 = \dfrac{v}{\lambda_1} - \dfrac{v}{\lambda_2} \Rightarrow 2 = \dfrac{v}{2} - \dfrac{v}{2.02}$

$2 = \dfrac{v}{202} \Rightarrow v = 404$

5. Ans (3)

$\dfrac{V}{4\ell} = 220\ \text{Hz}$



$\dfrac{3\ell}{4}$

$n' = 3\dfrac{V}{4 \times \frac{3\ell}{4}} = \dfrac{V}{\ell} = 880\ \text{Hz}$

6. Ans (2)

$(f_{app.})_{direct} = 165\ \text{Hz}$

$(f_{app.})_{Reflected} = 170\ \text{Hz}$

number of beats = 170 – 165 = 5 beats

24. Ans (3)

$W_{ext} = PE_2 - PE_1$

$= -\dfrac{3GM^2}{(2r)} - \dfrac{(-3GM^2)}{r}$

$= +\dfrac{3}{2}\dfrac{GM^2}{r}$

25. Ans (2)

$$\frac{-GMm}{R}+\frac{1}{2}m\left(\frac{3}{4}V_e\right)^2=-\frac{GMm}{R+h}+0$$

$$\frac{-GM}{R}+\frac{1}{2}\times\frac{9}{16}\times\frac{2GM}{R}=-\frac{GM}{R+h}$$

$$\frac{-7GM}{16R}=-\frac{GM}{R+h}$$

$7R+7h=16R$

$$h=\frac{9R}{7}$$

26. Ans (1)

$$g=\frac{4}{3}\pi R\rho G$$

$g\propto R$

27. Ans (1)

A geostationary satellite has a time period of T= 24 hr.

From Kepler's Law,

$$T^2\propto R^3\Rightarrow\frac{T_2^2}{T_1^2}=\frac{R_2^3}{R_1^3}$$

$$\Rightarrow\frac{T_2^2}{(24)^2}=\frac{(R+2.5R)^3}{(R+6R)^3}$$

$$T=\sqrt{\frac{24\times 24}{8}}=6\sqrt{2}\ \text{hour}$$

28. Ans (3)

$$\frac{d}{R}\times 100=40$$

$$\frac{d}{R}=\frac{40}{100}$$

$$d=\frac{4R}{10}=\frac{4\times 6400}{10}=2560\ \text{Km}$$

29. Ans (3)

For a satellite in an elliptical path, angular momentum and mechanical energy are conserved.

30. Ans (2)

$$V_i=\frac{30}{2}=15\ \text{m/s}$$

$$V_f=\frac{30}{3}=10\ \text{m/s}$$

Impulse $=\overrightarrow{\Delta p}$

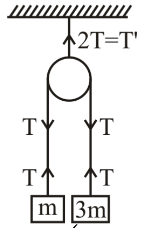
$=m\overrightarrow{\Delta v}$

$$\frac{20}{1000}(-10-15)$$

$$=-\frac{500}{1000}$$

$=-0.5$ kg m/s

31. Ans (2)



$$T=\left(\frac{2m_1m_2}{m_1+m_2}\right)g$$

$$T=\left(\frac{2(3m)(m)}{4m}\right)g$$

$$T=\frac{3mg}{2}$$

So $T'=2T=3mg$

32. Ans (4)

$$T_{max}=\frac{75}{100}Mg=\frac{3Mg}{4}$$

$\Rightarrow Mg-T=Ma$

$$\Rightarrow Mg-\frac{3Mg}{4}=Ma$$

$$\Rightarrow a\geq\frac{g}{4}$$

33. Ans (1)

In free falling condition tension is all strings become zero.

34. Ans (4)

$$a=\frac{F_{net}}{\text{total mass}}=\frac{20-10}{10}=1\ \text{m/s}^2$$

For 6 kg block

$20-T=6a$

$20-T=6\Rightarrow T=14N$

35. Ans (2)

Statement-II is incorrect because under zero net force block can move with constant velocity also.

SECTION-B

36. Ans (3)

$$n'=n\left(\frac{V}{V-V_S\cos 60^o}\right)$$

$$=100\left(\frac{330}{330-19.4\times\frac{1}{2}}\right)$$

$$=100\times\frac{330}{320.3}$$

$=100\times 1.03$

$=103$ Hz

37. Ans (3)

$\omega_1 = 1000\ \pi$, $\omega_2 = 998\ \pi$

number of beats $= \dfrac{\omega_1 - \omega_2}{2\pi}$

$= \dfrac{1000\pi - 998\pi}{2\pi} = 1$

38. Ans (4)

Frequency COP, $n_N = (2N+1)\dfrac{v}{4L}$

for $N = 0$, $n_0 = 100$ Hz

$n = 1$, $n_1 = 300$ Hz

$n = 2$, $n_2 = 500$ Hz

$n = 3$, $n_3 = 700$ Hz

$n = 4$, $n_4 = 900$ Hz

$n = 5$, $n_5 = 1100$ Hz Total = 6

Which are less than 1250 Hz.

46. Ans (1)

$V_{Centre} = V_{Solid} + V_{Shell}$

$= -\dfrac{3GM}{2R} - \dfrac{GM}{2R}$

$= -\dfrac{4GM}{2R} = -\dfrac{2GM}{R}$

47. Ans (4)

$V = x^2y$

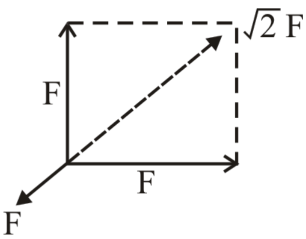
$\dfrac{\partial V}{\partial X} = 2xy \quad \dfrac{\partial V}{\partial y} = x^2$

$\vec{I} = (-2xy)\hat{i} - (x^2)\hat{j}$

$\vec{I}_{(2,1)} = -4\hat{i} - 4\hat{j}$

$|\vec{I}| = 4\sqrt{2}.\dfrac{N}{Kg}$

48. Ans (1)



Given that

$a = \dfrac{F}{m}$

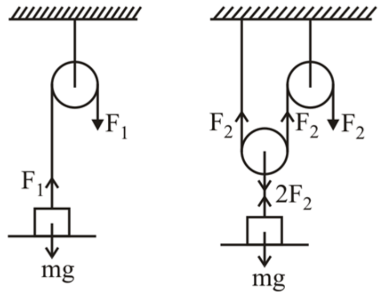
$F_{net} = \sqrt{2}F - F$

So $a_{body} = \dfrac{F_{net}}{m}$

$= (\sqrt{2}-1)\dfrac{F}{M}$

$= (\sqrt{2}-1)a$

49. Ans (1)



$F_1 = mg \qquad 2F_2 = mg$

$F_2 = \dfrac{mg}{2}$

means $F_1 > F_2$

50. Ans (2)

$\vec{p}_i = mV$

$\vec{p}_f = -mV$

$|\Delta\vec{p}| = |\vec{p}_f - \vec{p}_i|$

$|\Delta\vec{p}| = 2mV$

$\Delta t = \dfrac{\text{Dist}}{\text{Speed}} = \dfrac{\pi R}{V}$

So $F_{avg} = \dfrac{\Delta p}{\Delta t} = \dfrac{2mV^2}{\pi R}$

SUBJECT : CHEMISTRY

SECTION-A

55. Ans (4)

Calcium oxide is used as a flux and silicon dioxide is present as impurity, Lime stone is not used as acidic flux.

57. Ans (2)

$\boxed{\text{Stability} \propto \text{Bond order}}$

	O_2^+	O_2	O_2^-	O_2^{-2}
B.O.	2.5	2.0	1.5	1.0

SECTION-B

87. Ans (2)

at higher temp. reaction would be feasible

SUBJECT : BIOLOGY-I

SECTION-A

112. Ans (3)

NCERT Page # (E) 155, (H) 167

114. Ans (1)

NCERT XII Page No. # 159 (E), 174 (H)

116. Ans (3)

NCERT XII Pg. # 199 fig-11.4

120. Ans (1)

NCERT Pg # 198, 197

124. Ans (1)

NCERT (XIIth) Pg. # 81

SECTION-B

137. Ans (3)

NCERT XII, Pg.No. 152 Para-8.2.4

140. Ans (4)

(Normally $\longrightarrow$ Occasionally)

Ans : Statement -I = I

Statement-II = C

In NCERT given statement is $\longrightarrow$ Benign tumors normally remain confined to their original location and do not spread to other parts of the body.

We changed *occasionally* instead of normally

So meaning is changed

144. Ans (2)

NCERT (XIIth) Pg. # 84, 5.11

SUBJECT : BIOLOGY-II

SECTION-A

163. Ans (2)

NCERT Pg.# 263,16.2

176. Ans (2)

NCERT (XI) Pg. # 294, 295

182. Ans (3)

NCERT Pg. # 294

183. Ans (1)

NCERT-Page No. # 297

SECTION-B

192. Ans (3)

NCERT, Page # 266

193. Ans (1)

NCERT XIth Pg.#258

195. Ans (3)

NCERT XI, Pg 264

197. Ans (1)

NCERT (XI) Pg. # 290